འབྲུག་རྒྱལ་འཛིན་གཙུག་ལག་སློབ་གྲྭ ། ། ། །

College of Science and Technology
Rinchending: Bhutan

DBS101
Database Systems Fundamentals
(SS2026)

Practical 1 Report

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Program: BE 1SWE
Date: 13-3-2026

RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorized group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

Aim

To learn and build practical understanding by exploring and working with at least two different RDBMS software systems

Theory

MySQL is a popular open-source Relational Database Management System (RDBMS) used to store and manage data in many websites and applications. It uses SQL to create databases, add, update, and retrieve data. MySQL is known for being reliable, fast, and easy to use, and it can handle large amounts of data and multiple users at the same time.

Oracle:

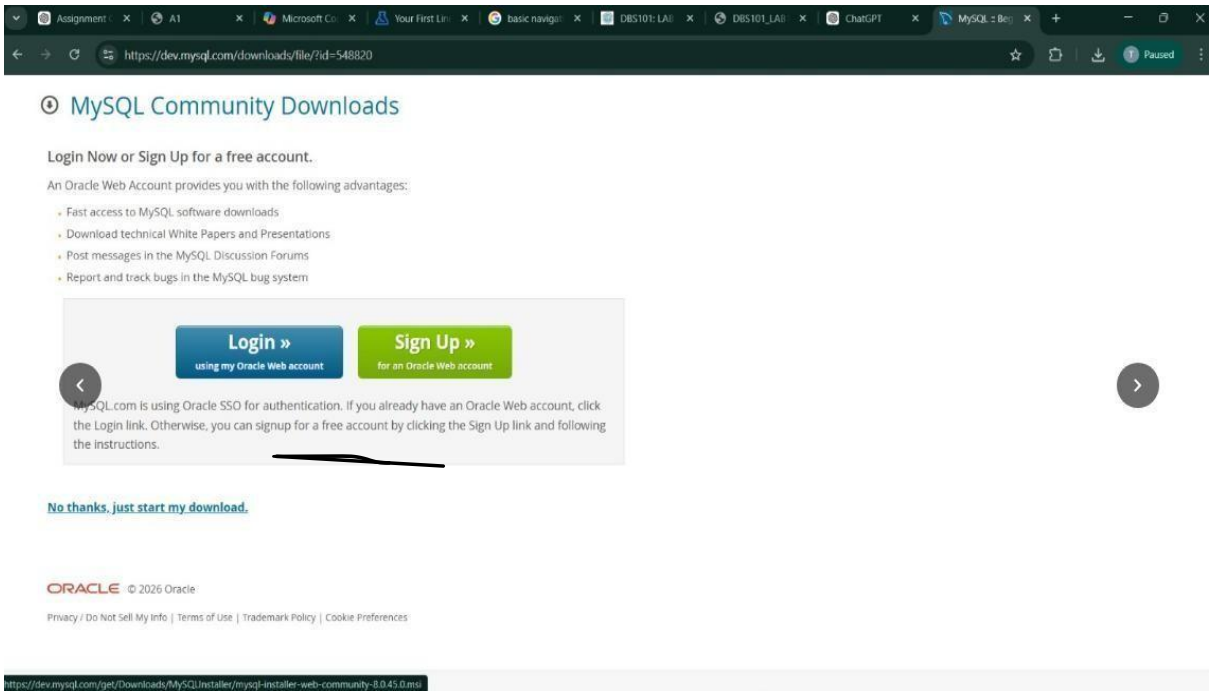
Oracle is a powerful Relational Database Management System developed by Oracle Corporation. It is commonly used by large organizations because of its advanced features, strong security, and high performance. Oracle also uses SQL and can manage very large and complex databases efficiently.

Implementation Steps

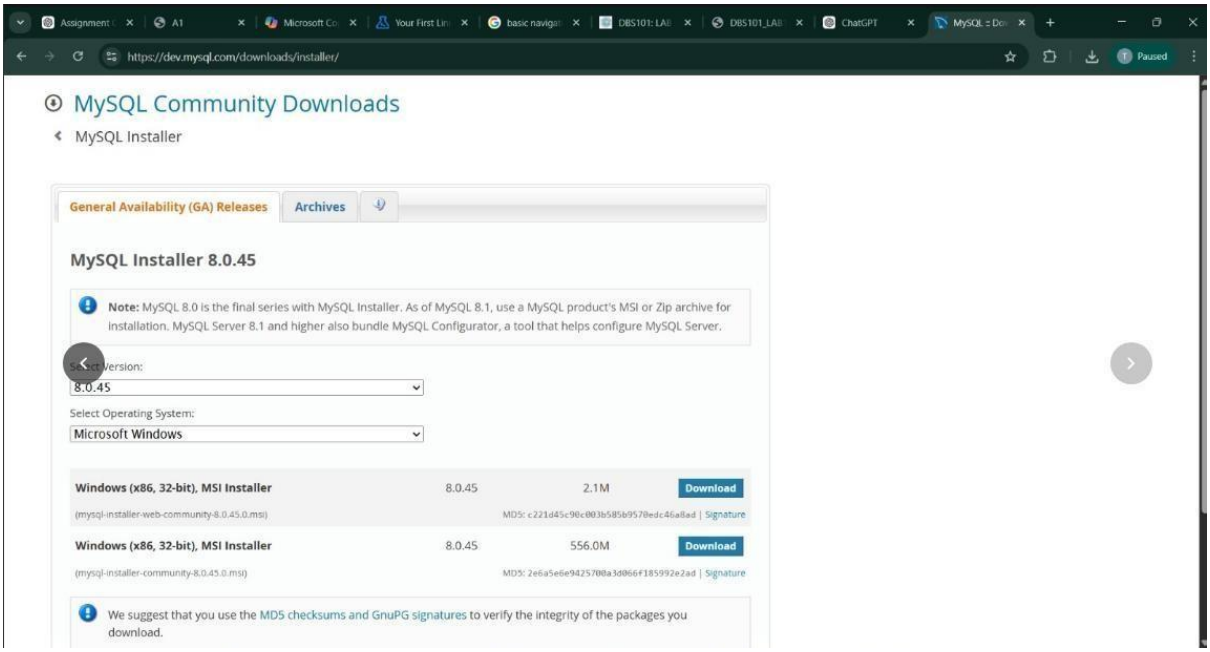
Work Evidence of Exercises; **add screenshots of completed tasks**

Example:

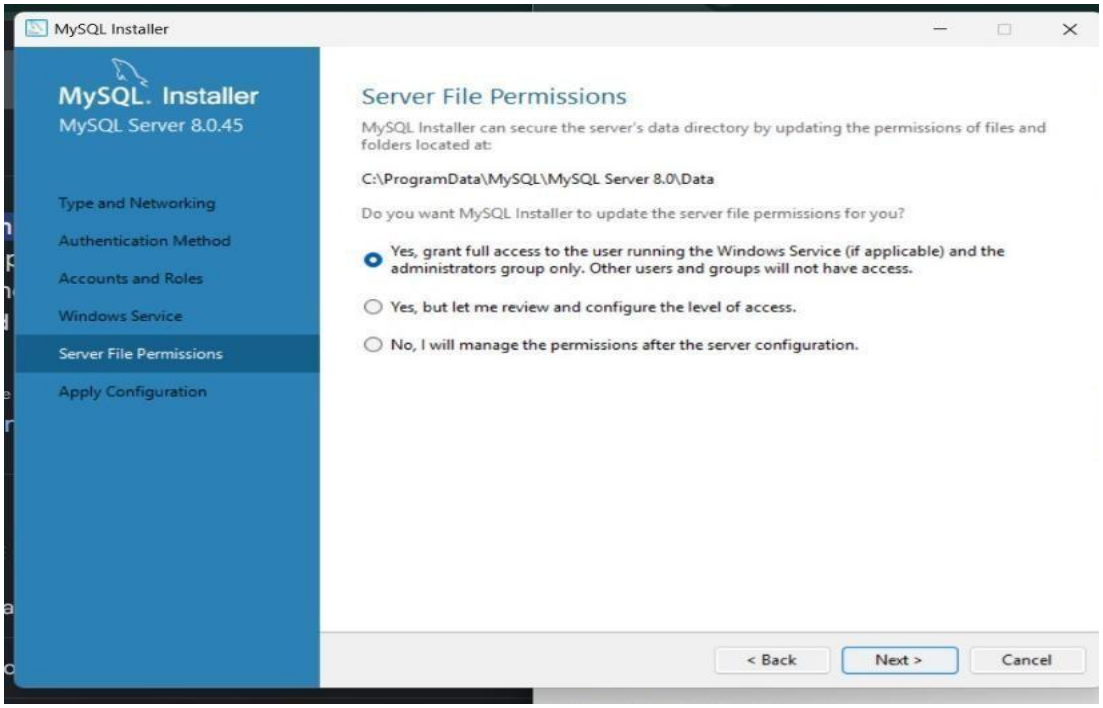
Task 1: MySQL Community Downloads



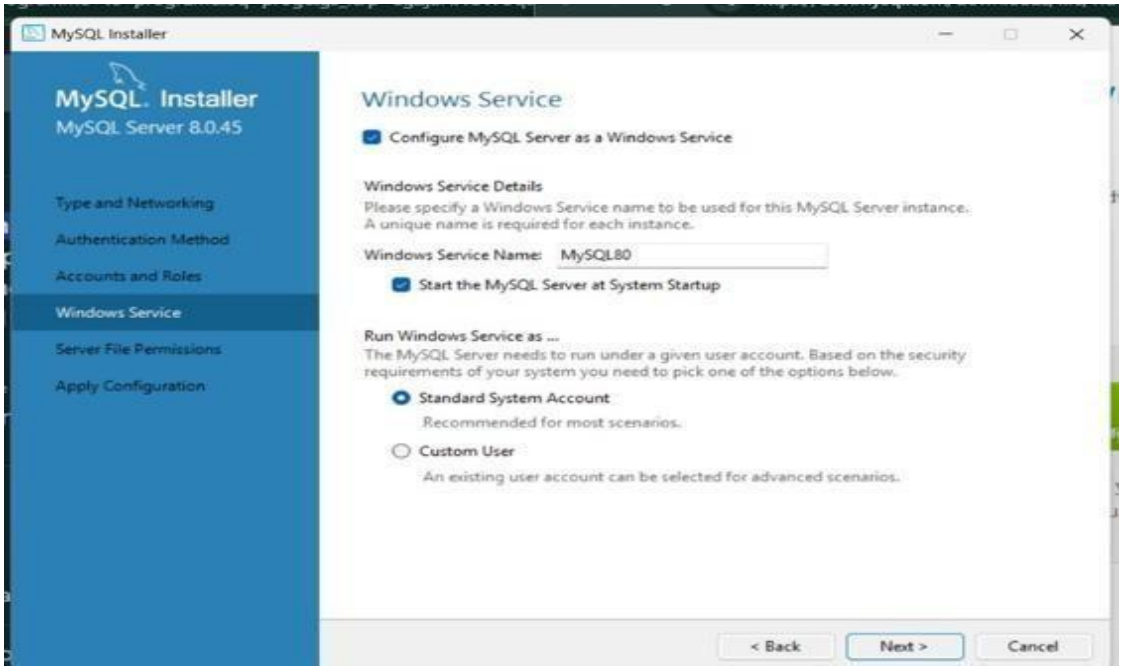
Task 2:MySQL Installer



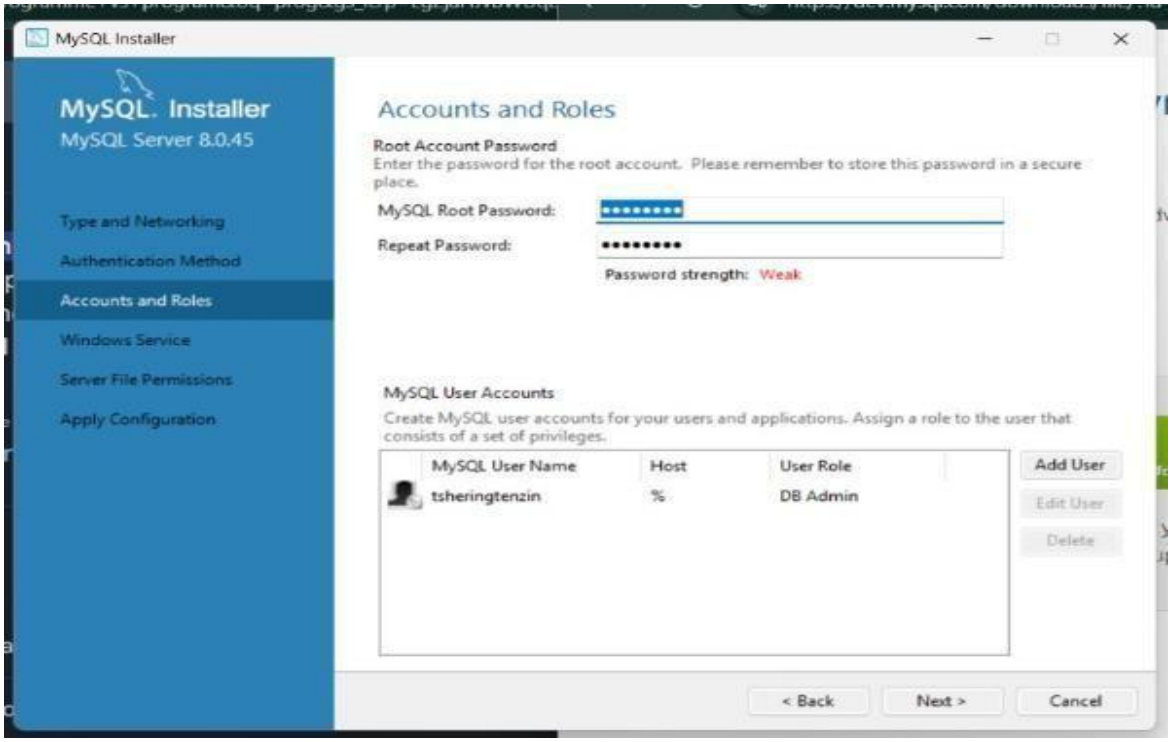
Task 3: server file permission



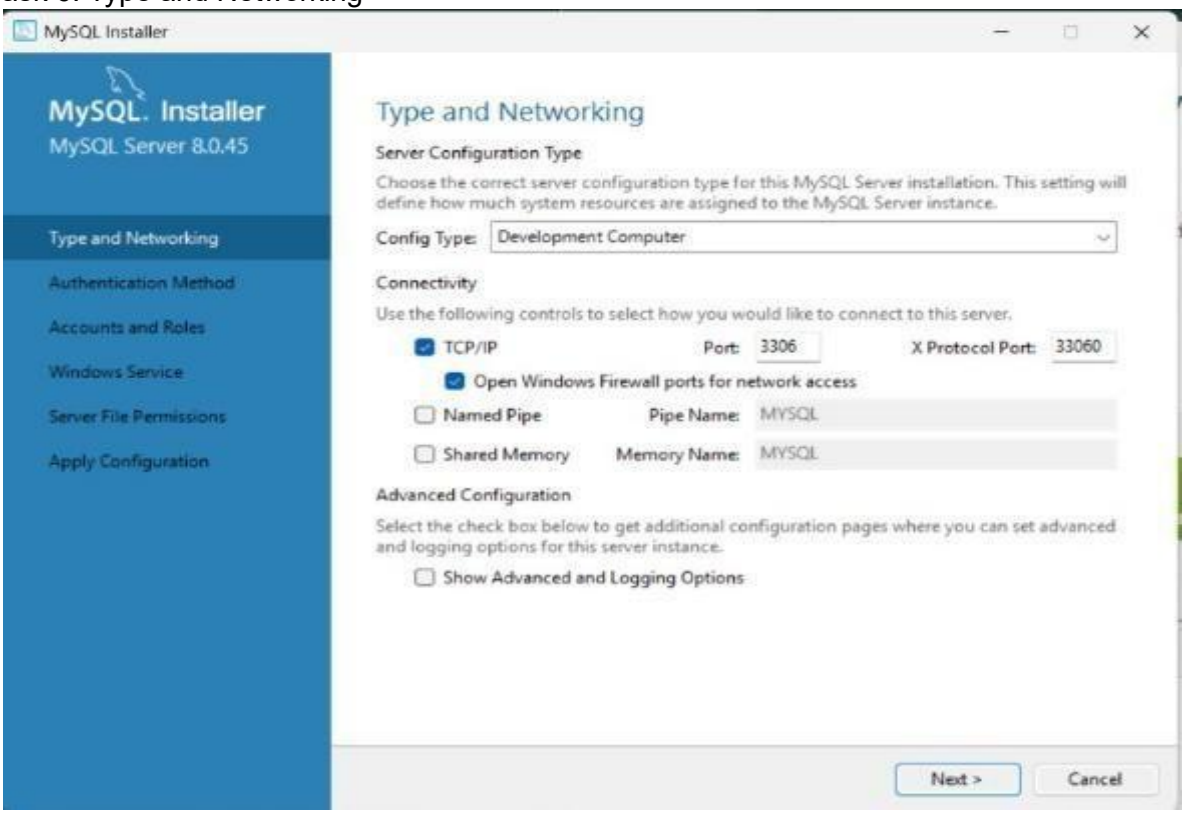
Task 4: Window Service



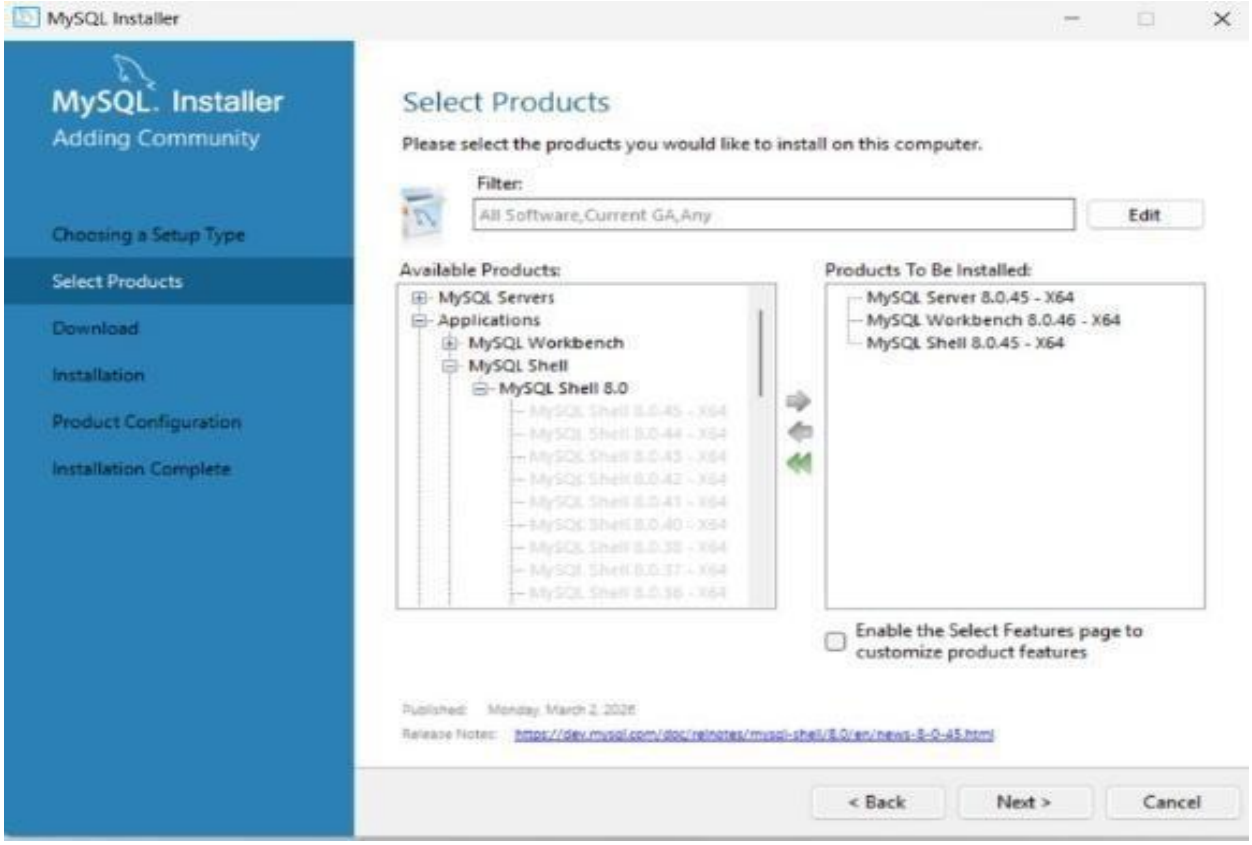
Task 5: Accounts and Roles



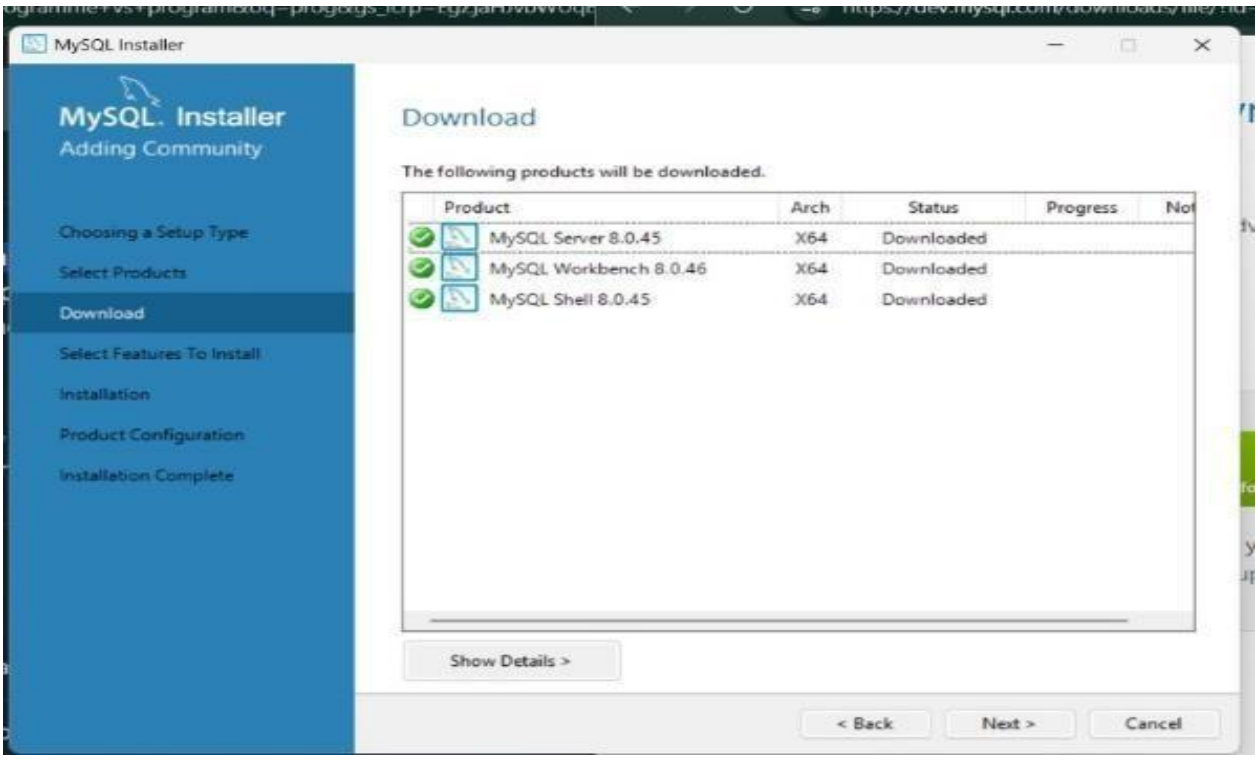
Task 6: Type and Networking



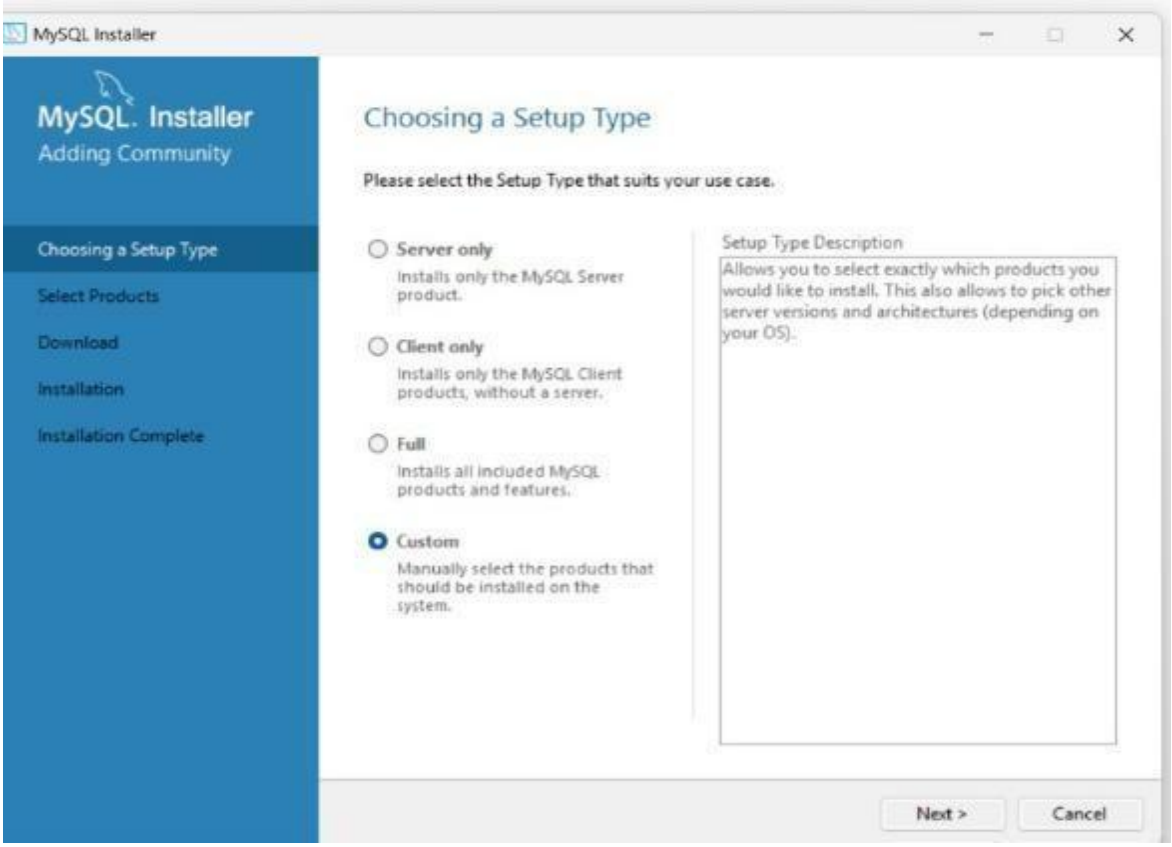
Task 7: Select Products



Task 8: Download



Task 9:



Task 10: Commanding on Command Prompt

```
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.28 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to
er version for the right syntax to use near 'database' at line 1
```

Conclusion

From this exercise, I learned the basic concepts of SQL and how it is used to create databases, manage tables, and retrieve data. The hands-on practice helped me understand how relational databases work and how SQL commands are applied in real situations. At the beginning, I faced some challenges such as syntax errors and remembering the correct commands. However, with practice and by trying different queries, I gradually improved my understanding. This exercise helped me build more confidence in working with databases and using SQL.

Reference

Oracle Corporation. (n.d.). Oracle Database documentation. https://www.oracle.com/database/

MySQL. (n.d.). MySQL documentation. https://dev.mysql.com/doc/

Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020). Database system concepts (7th ed.). McGraw-Hill Education.

Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems (7th ed.). Pearson.